

Practical 6: Hadoop Installation & HDFS

Install Java JDK 1.8

Java is required because Hadoop runs on the JVM.

Download and extract Hadoop

Places Hadoop files on your system.

Set JAVA_HOME and HADOOP_HOME

Makes Java and Hadoop accessible from Command Prompt.

Configure core-site.xml

Sets the default HDFS address.

Configure mapred-site.xml

Enables MapReduce to use YARN.

Create data/namenode and data/datanode

Creates storage locations for HDFS metadata and blocks.

Configure hdfs-site.xml

Defines replication and storage directories.

Configure yarn-site.xml

Sets YARN ResourceManager and NodeManager properties.

Edit hadoop-env.cmd

Points Hadoop to the installed JDK.

Install VC++ Redistributable and winutils

Provides Windows compatibility.

Format the NameNode

Initializes the HDFS filesystem.

Start Hadoop services

Starts HDFS and YARN daemons.

Verify using jps and localhost:9870

Confirms Hadoop is running correctly.

Practical 7: MapReduce Word Count

Start DFS and YARN

Starts Hadoop services.

Leave safe mode

Allows HDFS write operations.

Create /input_dir

Creates an input directory in HDFS.

Create input_file.txt

Prepares sample input text.

Upload the file

Copies the file from local storage to HDFS.

Verify the upload

Checks that the file exists and contents are correct.

Run the WordCount program

Executes the MapReduce example.

Check output directory

Confirms output files were created.

Display output

Shows the final word-frequency result.